CS548T 02. Parameter estimation

To find a probability dist. of r.v. X.

(1) choose a parameter model (Parameter θ)

(2) assemble a collection of samples of X

$D = \{x_1 \cdots x_n\}$ ↓ observation.

- assume x_i's are iid samples <u>indenpently</u> (independently identically distributed)

- Maximum Likelihood principle:

" the optimal parameter θ^* is that which maximizes the likelihood of observing the data D."

⇓

$$\theta^* = \arg\max_\theta P(D|\theta)$$ = likelihood of D with parameter θ

$$= \arg\max_\theta \log P(D|\theta)$$

or

$$= \arg\min_\theta -\log P(D|\theta)$$

observation D:

$$P(D|\theta) = P(x_1 \cdots x_N|\theta)$$ → iid.

$$= P(x_1|\theta) \cdots P(x_N|\theta)$$

$$= \prod_{i=1}^{N} P(x_i|\theta)$$

$$\Rightarrow \theta^* = \arg\max_\theta P(D|\theta)$$

$$= \arg\max_\theta \log P(D|\theta)$$

$$= \arg\max_\theta \sum_{i=1}^{N} \log P(x_i|\theta)$$

- $\sum_{i=1}^{N} \log P(x_i|\theta)$: log-likelihood term.

⇓

- $L = \log P(D|\theta) = \sum_{i=1}^{N} \log P(x_i|\theta)$

Solve (~~Sola~~ Scalar θ)

① $\frac{\partial L}{\partial \theta} = 0$ at θ^*

② $\frac{\partial^2 L}{\partial \theta^2} < 0$ at θ^*

② check

For θ is a vector.

① $\nabla_\theta L(\theta) = \begin{bmatrix} \frac{\partial L}{\partial \theta_1} \\ \vdots \\ \frac{\partial L}{\partial \theta_p} \end{bmatrix}$: gradient is 0.

② $\nabla_\theta^2 L(\theta) = \begin{bmatrix} \frac{\partial^2 L}{\partial \theta_1^2} & \cdots & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_p} \\ & \ddots & \vdots \\ & & \frac{\partial^2 L}{\partial \theta_p^2} \end{bmatrix} \prec 0$ • Negative definite. Hessian matrix

(③ ↗ $H < 0$ if $\forall \theta.\ \theta^T H \theta < 0$)

Bernoulli dist.

$\theta = \pi.\quad 0 \le \pi \le 1 \qquad P(x) = \pi^x (1-\pi)^{1-x} \qquad x \in \{0,1\}.$

$$L(\theta) = \sum_{i=1}^{N} \log P(x_i|\theta)$$

$$= \sum_{i=1}^{N} \log \theta^{x_i}(1-\theta)^{1-x_i}$$

$$= \sum_{i=1}^{N} (x_i \log\theta + (1-x_i)\log(1-\theta))$$

$$= \sum_{i=1}^{N} x_i \cdot \log\theta + \left(N - \sum_{i=1}^{N} x_i\right)\log(1-\theta)$$

$$\frac{\partial L}{\partial \theta} = \sum_{i=1}^{N} x_i \cdot \frac{1}{\theta} + \left(N - \sum_{i=1}^{N} x_i\right)\frac{-1}{1-\theta} = 0$$

$$\sum_{i=1}^{N} x_i (1-\theta) - \left(N - \sum_{i=1}^{N} x_i\right)\theta = 0$$

$$\hat{\theta} = \frac{1}{N}\sum_{i=1}^{N} x_i$$ → sample mean

$$\frac{\partial^2 L}{\partial \theta^2} = -\frac{\sum_{i=1}^{N} x_i}{\theta^2} + \frac{\left(\sum_{i=1}^{N} x_i - N\right)}{(1-\theta)^2} \le 0 \quad (x_i \in \{0,1\}).$$

$$\Rightarrow \theta^* = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Gaussian: $P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$

parameters $\mu.\ \sigma^2$.

σ^2 is known: $\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i$

μ is known: $\hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)^2$ (sample variance)

⇓ Multivariant Gaussian.

- $\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i$
- $\hat{\Sigma} = \frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)(x_i - \mu)^T$

estimate vs estimator.

estimate is a value ($\hat{\mu} = \sum_{i=1}^{N} x_i$)

estimator is a r.v. ($\mu_{ML} = \sum_{i=1}^{N} X_i$)

- Bias =

$Bias(\hat{\theta}) = E[\hat{\theta} - \theta] = E[\hat{\theta}] - \theta$

- Variance:

$Var(\hat{\theta}) = E_{x_1 \cdots x_N}[\hat{\theta} - E[\hat{\theta}]]$

$$
\begin{aligned}
Bias(\hat{\mu}) &= E[\hat{\mu}] - \mu \\
&= \mu - \mu \\
&= 0
\end{aligned}
$$

$Var(\hat{\mu}) = \frac{1}{N}\sigma^2$

$E[\hat{\sigma}^2] = \frac{N-1}{N}\sigma^2 = (1-\frac{1}{N})\sigma^2.$

$Bias(\hat{\sigma}^2) = \frac{-1}{N}\sigma^2$

$\Downarrow$

make it unbias

$\hat{\hat{\sigma}}^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2$

- Least Square regression:

(Supervised Learning)

$D = \{(x_i\ y_i)\}_{i=1}^{I}$

$f(x_0|\theta) = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots \theta_p x^p$

$\Downarrow$

$f(x) = \phi(x)^T\theta$

observe a noise $\varepsilon \sim N(0, \sigma^2)$

$y = f(x|\theta) + \varepsilon$

$p(y|x, \theta) = N(y|f(x, \theta), \sigma^2)$

$\Downarrow$

$$
\begin{aligned}
\theta^* &= \arg\max \sum \log N(y_i | f(x_i, \theta), \sigma^2) \\
&\vdots \\
&= \arg\min \|y - \Phi^T\theta\|^2 \\
&= (\Phi\Phi^T)^{-1}\Phi y
\end{aligned}
$$

$\Phi = [\phi(x_1) \cdots \phi(x_N)]$